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September 19, 2026

1 Introduction

L^AT_EX has many caveats, you should search stack overflow for latex tips whenever you feel something looks bad, for instance: When “ quoting ”, should be used instead of ”. For example, “test” vs ”test”.

Please refer to the spec, the word count and page count is strict. Feel free to change the section headings (and we recommend you do).

Always remember to cite anything that is not your work. For instance, you must cite the datasets like this [1, 2].

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2 Preprocessing

The TLC receives trip records from the technology providers in each vehicle and makes no promise that they are accurate [3], so every record is checked before use. All checks, cut-offs and removals are run on the full data in PySpark, one month at a time. Table 1 gives the shape of the data after each step, and the points below give the reason for each step.

Table 1: Shape of the taxi data after each cleaning step. *Removed* is the share of raw rows removed by that step; – marks a step that does not apply to the service.

Step	Yellow taxi			HVFHV (Uber and Lyft)		
	Rows	Cols	Removed	Rows	Cols	Removed
Raw files (Jan 2023–Dec 2025)	128,202,548	19–20		715,550,152	24–25	
1 Keep used columns	128,202,548	14	0%	715,550,152	10	0%
2 Pickup in the file’s month	128,201,184	14	<0.01%	715,550,152	10	0%
3 Drop-off after pickup	127,625,804	14	0.45%	715,519,470	10	<0.01%
4 Fare collected (payment type)	124,407,631	14	2.51%	–		
5 Positive fare	123,959,890	14	0.35%	–		
6 Known rate code	122,483,608	14	1.15%	–		
7 Positive driver pay	–			714,346,062	10	0.16%
8 Positive distance	120,019,973	14	1.92%	714,260,147	10	0.01%
9 Outlier caps	119,649,902	14	0.29%	714,147,203	10	0.02%
10 Known pickup and drop-off zones	118,440,009	14	0.94%	683,572,826	10	4.27%
Cleaned (share of raw rows kept)	118,440,009 × 14 (92.4%)			683,572,826 × 10 (95.5%)		

2.1 Taxi trip data

- **One schema (step 1).** Column types and names change during 2023, so each month is cast to one schema before stacking, keeping only the columns used later. The extra raw column in 2025 records the new toll fee.
- **Nulls are kept.** Yellow nulls fall in four columns only, and in every month on exactly the Flex Fare trips (payment type 0, a price agreed upfront [4]). These are real trips that grew from 3.4% of yellow rows in 2023 to 23.8% in 2025, so dropping them would fake a fall in trips after the toll. None of the four columns is used, so nothing is imputed. The toll fee column is null only before the toll began.
- **Impossible times (steps 2–3).** A pickup outside the file’s month has a wrong date, and a drop-off at or before pickup gives no trip length. 93% of the yellow rows removed at step 3 come from vendor 7 (Helix), new in December 2024 [4], which records this on every trip. Its pickups are in the tolled area as often as yellow pickups overall (56.4% and 56.8%), so removing it does not favour either side of the comparison.
- **Unreliable fares (steps 4–6, yellow).** Payment types 3, 4 and 6 are no charge, dispute and voided trips, and rate code 99 is unknown [4], so their revenue cannot be trusted. Most zero or negative fares are reversals: 88–92% of negative cash, no-charge and dispute fares cancel another row exactly. Flex Fare trips skip steps 5–6, because their negative fares (20.5% of Vendor 2’s in 2025) cancel nothing. They are kept with unknown revenue, so 2025 yellow earnings are known for 94.7–96.6% of trips in each zone group.
- **Unpaid trips (step 7, HVFHV).** Driver pay is what the driver earns after commission, excluding tips [5]. It is the Uber and Lyft earnings outcome, so zero or negative pay is removed. Most of these rows are from January–March 2023, before the comparison period.
- **Outliers (steps 8–9).** Zero distance cannot be a completed trip. The caps remove trips under 1 minute or over 5 hours, over 100 miles, over 60 mph on average, or paying over \$500. They were set from full-data histograms where each tail turns into impossible values, and sit well above every 99.9th percentile (under 2 hours, 54 miles, 48 mph and \$152).

- **Unknown zones (step 10).** Zones 264 (“Unknown”) and 265 (“Outside of NYC”) have no shape [6], so the trip cannot be placed relative to the tolled area.
- **Summary table.** Earnings are driver pay, or for yellow the fare plus the driver’s own surcharges, after removing the city and MTA charges that Vendor 1 adds to the same column. Trips are summed per service $\times$ pickup zone $\times$ drop-off zone group $\times$ date $\times$ time band, with medians taken over every trip and empty combinations kept as 0 trips: $11,529,920 \times 17 (2 \times 263 \times 5 \times 1,096 \times 4)$. This counts all 802 million cleaned trips in a table small enough to model, and its totals are checked against Table 1.

2.2 External data

- **Zone groups.** The MTA list of tolled zones (38×2) [7] is joined to the TLC zone lookup (265×4) on zone ID, with each zone’s distance to the tolled area from the TLC shapefile (263×7) [6]: 265×8 . Zones within 1 km form a *ring* (19 zones, 13 across a river), since riders can walk out of the tolled area and drivers can wait at its edge; as controls they would overstate the toll’s effect. Any cut-off from 0.94 to 1.05 km gives the same ring. The other 206 zones are controls. A trip is *treated* if either end is in the tolled area, otherwise *ring* if either end is in the ring.
- **Weather and holidays.** NOAA daily Central Park records [8], $1,096 \times 7$, already metric. The station ID is dropped and the one missing snow depth (17 September 2025, 21°C) is set to 0: $1,096 \times 6$. A flag marks the 35 US federal holidays of 2023–2025 [9], on the weekday observed.
- **Joins.** Both join on date: $11,529,920 \times 17 \rightarrow \times 22 \rightarrow \times 23$, with the row count checked unchanged. Both models are fitted to this table.

Does cleaning bias the comparison? A rule can bias it if it removes a larger share from one group after the toll. From 2024 to 2025 the share of yellow rows removed rose by 2.6 points for treated trips and 3.0 for control trips, a gap of at most 0.4 points. For HVFHV, step 10 removed 4.17% of treated rows in 2024 and 4.84% in 2025, and cannot remove control trips, so up to 0.7 points of any fall in treated HVFHV trips may come from trips to unknown zones.

3 Analysis and Geospatial Visualisation

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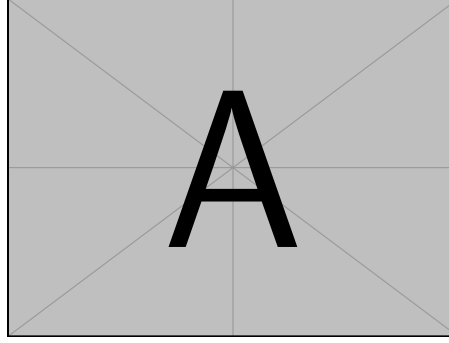


Figure 1: Some caption

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4 Modelling

Example of a maths equation:

$$Y = X\beta + \epsilon \quad (1)$$

Example of an aligned equation (& denotes the symbol to align):

$$\begin{aligned} E[\mathbf{y}] &= X\beta + E[0] \\ &= X\beta \end{aligned}$$

Example of an in-line equation $\epsilon \sim N(0, 1)$

5 Discussion

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6 Recommendations

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7 Conclusion

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References

- [1] City of Melbourne. *Microclimate Sensor Readings*. <https://data.melbourne.vic.gov.au/Environment/Microclimate-Sensor-Readings/u4vh-84j8>. Accessed: 2022-08-01.
- [2] City of Melbourne. *Microclimate Sensor Locations*. <https://data.melbourne.vic.gov.au/Environment/Microclimate-Sensor-Locations/irqv-hjr4>. Accessed: 2022-08-01.
- [3] NYC Taxi and Limousine Commission. *TLC Trip Record Data*. <https://www.nyc.gov/site/tlc/about/tlc-trip-record-data.page>. Accessed: 2026-09-15.
- [4] NYC Taxi and Limousine Commission. *Data Dictionary – Yellow Taxi Trip Records*. https://www.nyc.gov/assets/tlc/downloads/pdf/data_dictionary_trip_records_yellow.pdf. Accessed: 2026-09-15. Mar. 2025.
- [5] NYC Taxi and Limousine Commission. *Data Dictionary – High Volume FHV Trip Records*. https://www.nyc.gov/assets/tlc/downloads/pdf/data_dictionary_trip_records_hvfhs.pdf. Accessed: 2026-09-15. Mar. 2025.
- [6] NYC Taxi and Limousine Commission. *Taxi Zone Lookup Table and Taxi Zone Shapefile*. <https://www.nyc.gov/site/tlc/about/tlc-trip-record-data.page>. Accessed: 2026-09-15.
- [7] Metropolitan Transportation Authority. *MTA Central Business District Taxi Zones*. <https://data.ny.gov/d/yfdc-w5jh>. Accessed: 2026-09-15.
- [8] NOAA National Centers for Environmental Information. *Global Historical Climatology Network – Daily, Station USW00094728 (New York Central Park)*. <https://www.ncei.noaa.gov/access/services/data/v1?dataset=daily-summaries&stations=USW00094728>. Accessed: 2026-09-15.
- [9] U.S. Office of Personnel Management. *Federal Holidays*. <https://www.opm.gov/policy-data-oversight/pay-leave/federal-holidays/>. Accessed: 2026-09-19.